

LAB 2: SAT Constraint Closure and Coupled Field Dynamics

I. Dirac Constraint Closure (u^μ Sector)

Primary Constraint:

$$\phi_1 = u^\mu u_\mu + 1 = -u_0^2 + u_1^2 + 1$$

Secondary Constraint (time evolution + theta coupling):

$$\phi_2 = -2 u_0 \frac{du_0}{dt} + 2 u_1 \frac{du_1}{dt} + \beta * [(\frac{du_0}{dt} * u_0 + \frac{du_1}{dt} * u_1) * \frac{d\theta}{dt} + (\frac{du_0}{dx} * u_0 + \frac{du_1}{dx} * u_1) * \frac{d\theta}{dx}]$$

Tertiary Constraint:

ϕ_3 not required; ϕ_2 closes under time evolution.

ϕ_1 and ϕ_2 are second-class $\Rightarrow$ Use Dirac brackets for quantization.

II. Coupled theta-tau Field Theory (1+1D)

Lagrangian:

$$L = \frac{1}{2} \left(\frac{d\theta}{dt} \right)^2 - \frac{1}{2} \left(\frac{d\theta}{dx} \right)^2 - \mu^2 \cos(3\theta) - \epsilon_f \tau^2 - \epsilon_f \gamma \tau^2 \left(\frac{d\theta}{dx} \right)^2$$

Euler-Lagrange Equations:

EOM for theta:

$$\frac{d^2\theta}{dt^2} - \frac{d^2\theta}{dx^2} + 3\mu^2 \sin(3\theta) + 2\epsilon_f \gamma \tau^2 \frac{d^2\theta}{dx^2} + 4\epsilon_f \gamma \tau \left(\frac{d\tau}{dx} \right) \left(\frac{d\theta}{dx} \right) = 0$$

EOM for tau:

$$\frac{\delta L}{\delta \tau} = -2\epsilon_f (1 + \gamma \left(\frac{d\theta}{dx} \right)^2) \tau = 0$$